18. 04. 2017. K1

1. Начртати структурни блок дијаграм система чији је сигнал на излазу дат Z трансформацијом у облику:

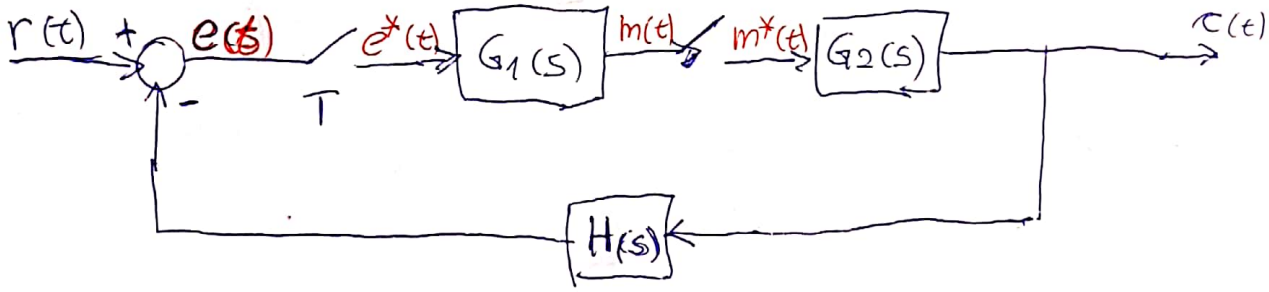
a) $C(z) = \dfrac{G_1(z) \cdot G_2(z) \cdot R(z)}{1 + G_1(z) \cdot G_2 \cdot H(z)}$

b) $C(z) = \dfrac{G_1 R(z) \cdot G_2(z)}{1 + G_1 G_2 H(z)}$

Улазни сигнал система је дат са $r(t)$.

РЈЕШЕЊЕ:

a)



$$\left.\begin{aligned} C(s) &= G_2(s) \cdot M^*(s) \\ M^*(s) &= G_1(s) \cdot E^*(s) \\ M^*(s) &= G_1^*(s) \cdot E^*(s) \end{aligned}\right\} \Rightarrow C(s) = G_1^*(s) \cdot G_2(s) \cdot E^*(s)$$

$$E(s) = R(s) - H(s) \cdot C(s) \xrightarrow{\text{ТАФДП}}$$

$$E(s) = R(s) - G_1^*(s) \cdot G_2(s) \cdot H(s) \cdot E^*(s) \xrightarrow{\text{ТАФДП}}$$

$$E^*(s) = R^*(s) - G_2 H^*(s) \cdot G_1^*(s) \cdot E^*(s) \quad / Z$$

$$E(z) = R(z) - G_2 H(z) \cdot G_1(z) \cdot E(z)$$

$$E(z) = \frac{R(z)}{1 + G_2 H(z) \cdot G_1(z)}$$

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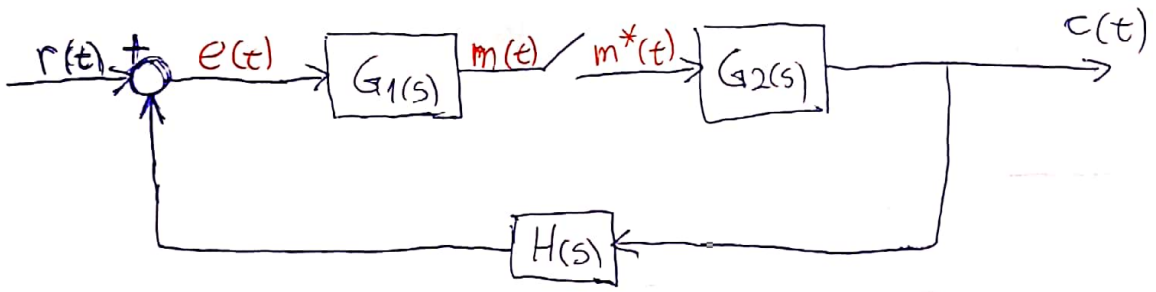
$$C(s) = G_1^*(s) \cdot G_2(s) \cdot E^*(s) \underset{\text{ТАФДП}}{\Longrightarrow} C^*(s) = G_1^*(s) \cdot G_2^*(s) \cdot E^*(s) \,/z$$

$$C(z) = G_1(z) \cdot G_2(z) \cdot E(z) = G_1(z) \cdot G_2(z) \cdot \frac{R(z)}{1 + G_1(z) \cdot G_2 H(z)}$$

$$\boxed{C(z) = \frac{G_1(z) \cdot G_2(z) \cdot R(z)}{1 + G_1(z) \cdot G_2 H(z)}}$$

b)



$$\cancel{C(s) = G_2(s) \cdot M^*(s)}$$

$$\cancel{M(s) = G_1(s) \cdot E(s) \underset{\text{ТАФДП}}{\Longrightarrow} M^*(s) = G_1 E^*(s)} \Big\} \cancel{\Rightarrow C(s) = G_2(s) \cdot G_1 E^*(s)}$$

$$\cancel{E(s) = R(s) - C(s) H(s) \underset{\text{ТАФДП}}{\Longrightarrow} E^*(s) = R^*(s) - H C^*(s)}$$

$$\cancel{C(s) = G_2(s) \cdot G_1 [R^*(s) - H C^*(s)]}$$

$$C(s) = G_2(s) \cdot M^*(s) \underset{\text{ТАФДП}}{\Longrightarrow} C^*(s) = G_2^*(s) \cdot M^*(s)$$

$$\left.\begin{aligned} M(s) &= G_1(s) \cdot E(s) \\ E(s) &= R(s) - H(s) \cdot C(s) \end{aligned}\right\} \Rightarrow M(s) = G_1(s) \cdot R(s) - G_1(s) \cdot H(s) \cdot C(s)$$

$$M(s) = G_1(s) \cdot R(s) - G_1(s) \cdot H(s) \cdot G_2(s) \cdot M^*(s) \underset{\text{ТАФДП}}{\Longrightarrow}$$

$$M^*(s) = G_1 R^*(s) - G_1 G_2 H^*(s) \cdot M^*(s) \Rightarrow M^*(s) = \frac{G_1 R^*(s)}{1 + G_1 G_2 H^*(s)}$$

$$C^*(s) = G_2^*(s) \cdot \frac{G_1 R^*(s)}{1 + G_1 G_2 H^*(s)} \,/z \Rightarrow \boxed{C(z) = \frac{G_2(z) \cdot G_1 R(z)}{1 + G_1 G_2 H(z)}}$$

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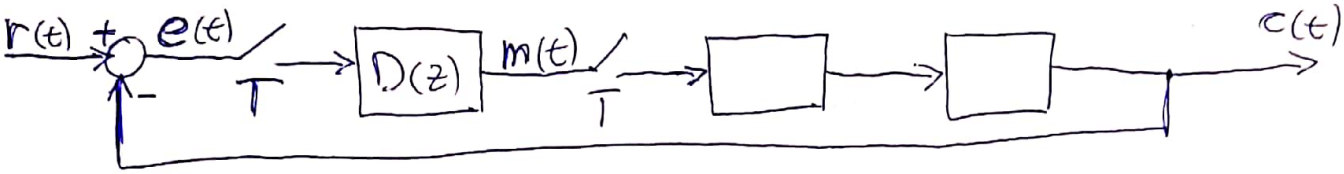
② Дигитални систем управљања дат је на слици.



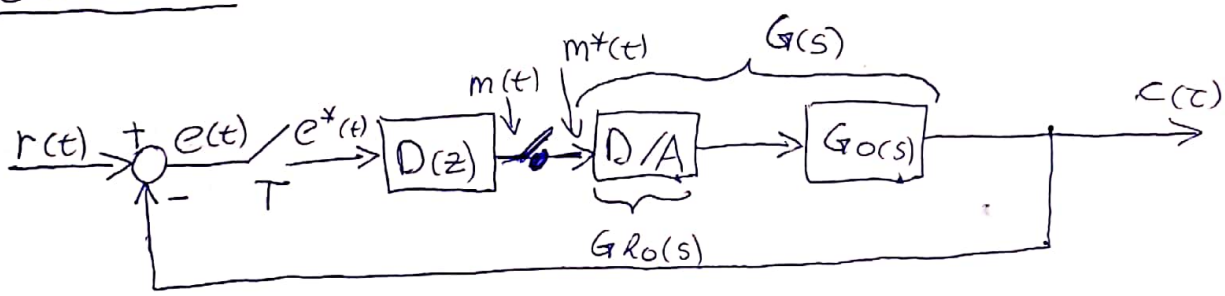
a) Који блок на слици треба да представља D/A конвертор, а који објекат управљања?

b) Одредити ф-ју преноса у Z домену ако дигитални блок D(z) ради по алгоритму $m(kT)=e(kT)$ и ако је објекат управљања представљен са $G_0(s)=\frac{1}{s(s+1)}$. Периода одмјеравања је $T=1[s]$.

c) Одредити и скицирати јединични одскочни одзив система у Z домену за првих 10 [s].

РЈЕШЕЊЕ:

a)



b)

$$W(z)=\frac{C(z)}{R(z)}$$

$$C(s)=G(s)\cdot M^*(s) \xrightarrow[\text{ТАФДП}]{} C^*(s)=G^*(s)\cdot M^*(s)$$

$$M(s)=D(z)\cdot E^*(s) \xrightarrow[\text{ТАФДП}]{} M^*(s)=D(z)\cdot E^*(s)$$

$$E(s)=R(s)-C(s) \xrightarrow[\text{ТАФДП}]{} E^*(s)=R^*(s)-C^*(s)$$

$$M^*(s)=D(z)\,R^*(s)-D(z)\,C^*(s)$$

$$C^*(s)=G^*(s)\cdot D(z)\cdot R^*(s)-G^*(s)\cdot D(z)\cdot R^*(s) \quad /Z$$

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$$C(z)\left[1+G(z)\cdot D(z)\right]=G(z)\cdot D(z)\cdot R(z)$$

$$C(z)=\frac{G(z)\cdot D(z)}{1+G(z)\cdot D(z)}\cdot R(z)$$

$$W(z)=\frac{C(z)}{R(z)}=\frac{G(z)\cdot D(z)}{1+G(z)\cdot D(z)}$$

$$\left.\begin{array}{l} M(z)=D(z)\cdot E(z) \\ D(z):\ m(kT)=e(kT)\ /Z \Rightarrow M(z)=E(z) \end{array}\right\} \Rightarrow D(z)=1$$

$$W(z)=\frac{G(z)}{1+G(z)}$$

$$G(z)=G_{R_0}G_g(z)=Z\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s(s+1)}\right\}=$$

$$=(1-z^{-1})\,Z\left\{\frac{1}{s^2(s+1)}\right\}$$

$$\frac{1}{s^2(s+1)}=\frac{A}{s^2}+\frac{B}{s}+\frac{C}{s+1}=\frac{As+A+Bs^2+Bs+Cs^2}{s^2(s+1)}$$

$$\left.\begin{array}{l} B+C=0 \\ A+B=0 \\ A=1 \end{array}\right\} \Rightarrow A=1,\ B=-1,\ C=1$$

$$G(z)=\frac{z-1}{z}\cdot Z\left\{\frac{1}{s^2}-\frac{1}{s}+\frac{1}{s+1}\right\}=\frac{z-1}{z}\cdot\left[\frac{Tz}{(z-1)^2}-\frac{z}{z-1}+\frac{z}{z-e^{-T}}\right]$$

$T=1[s]$

$$G(z)=\frac{z-1}{z}\cdot\left[\frac{z}{(z-1)^2}-\frac{z}{z-1}+\frac{z}{z-e^{-1}}\right]=\frac{1}{z-1}-1+\frac{z-1}{z-0.3679}$$

$$G(z)=\frac{z-0.3679\cancel{-z^2}+1.3679z-0.3679\cancel{+z^2}-2z+1}{(z-1)(z-0.3679)}$$

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$$G(z) = \frac{0.3679z + 0.2642}{(z-1)(z-0.3679)}$$

$$W(z) = \frac{G(z)}{1+G(z)} = \frac{\dfrac{0.3679z + 0.2642}{(z-1)(z-0.3679)}}{1 + \dfrac{0.3679z + 0.2642}{(z-1)(z-0.3679)}}$$

$$W(z) = \frac{\dfrac{0.3679z + 0.2642}{\cancel{(z-1)(0.3679)}}}{\dfrac{z^2 - 1.3679z + 0.3679 + 0.3679z + 0.2642}{\cancel{(z-1)(z-0.3679)}}}$$

$$\boxed{W(z) = \frac{0.3679z + 0.2642}{z^2 - z + 0.6321}}$$

c) $R(z) = \dfrac{z}{z-1}$

$$C(z) = W(z) \cdot R(z) = \frac{0.3679z + 0.2642}{z^2 - z + 0.6321} \cdot \frac{z}{z-1}$$

$$C(z) = \frac{0.3679z^2 + 0.2642z}{z^3 - z^2 - z^2 + z + 0.6321z - 0.6321}$$

$$C(z) = \frac{0.3679z^2 + 0.2642z}{z^3 - 2z^2 + 1.6321z - 0.6321}$$

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$$(0.3679z^2+0.2642z):(z^3-2z^2+1.6321z-0.6321)=0.3679z^{-1}+z^{-2}+1.3996z^{-3}+1.3996z^{-4}\ldots$$

$$-(0.3679z^2-0.7358z+0.6004-0.2325z^{-1})$$

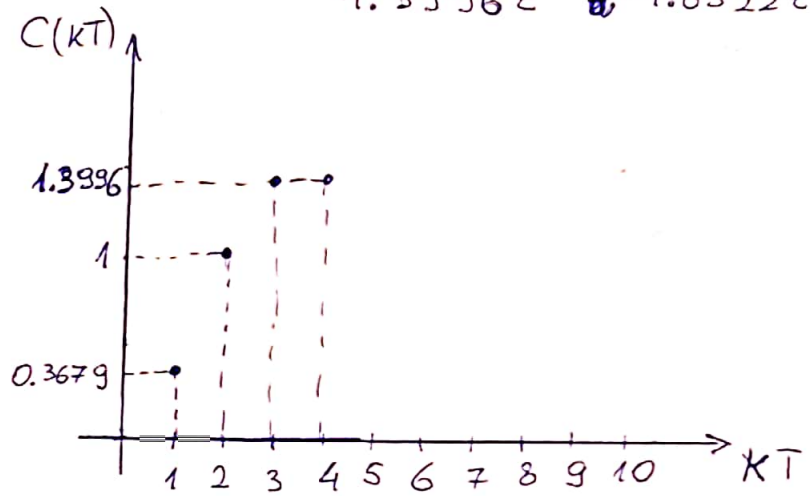
$$z-0.6004+0.2325z^{-1}$$

$$-(z-2+1.6321z^{-1}-0.6321z^{-2})$$

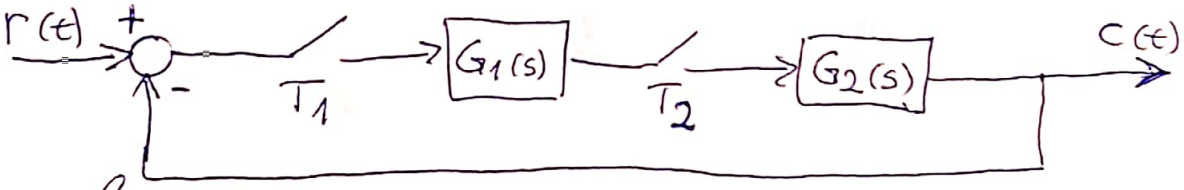
$$1.3996-1.3996z^{-1}+0.6321z^{-2}$$

$$-(1.3996-2.7992z^{-1}+2.2843z^{-2}-0.8847z^{-3})$$

$$1.3996z^{-1}-1.6522z^{-2}+0.8847z^{-3}$$



3. За систем приказан на ~~слици~~ слици одредити Z трансформацију сигнала на излазу.



Блокови $G_1(s)$ и $G_2(s)$ су описани са $G_1(s)=\frac{1}{s}$, $G_2(s)=\frac{1}{s+1}$. Усвојити да је на улазу система јединични одскочни сигнал и да су периоди одмјеравања $T_1=2[s]$ и $T_2=1[s]$.

РЈЕШЕЊЕ:

- Систем који је приказан на слици садржи одабирање различитим брзинама одабирања па га је потребно трансформисати да садржи прекидаче

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са идентичним периодима одабирања. Ово је могуће урадити само ако се почетне периоде одабирања односе као цијели бројеви $\frac{T_1}{T_2} = \frac{m}{n}$. У том случају се бира заједничка периода одабирања (сторија). Идеја лежи у чињеници да се одбирци сигнала одабирани са периодом $\frac{T}{n}$ могу добити као сума n низова одбирака тог истог сигнала који се одабирају са периодом T.

$$f^*(t) = f_1^*(t) + f_2^*\left(t + \frac{T}{n}\right) + f_3^*\left(t + \frac{2T}{n}\right) + \ldots + f_n^*\left(t + \frac{(n-1)T}{n}\right)$$

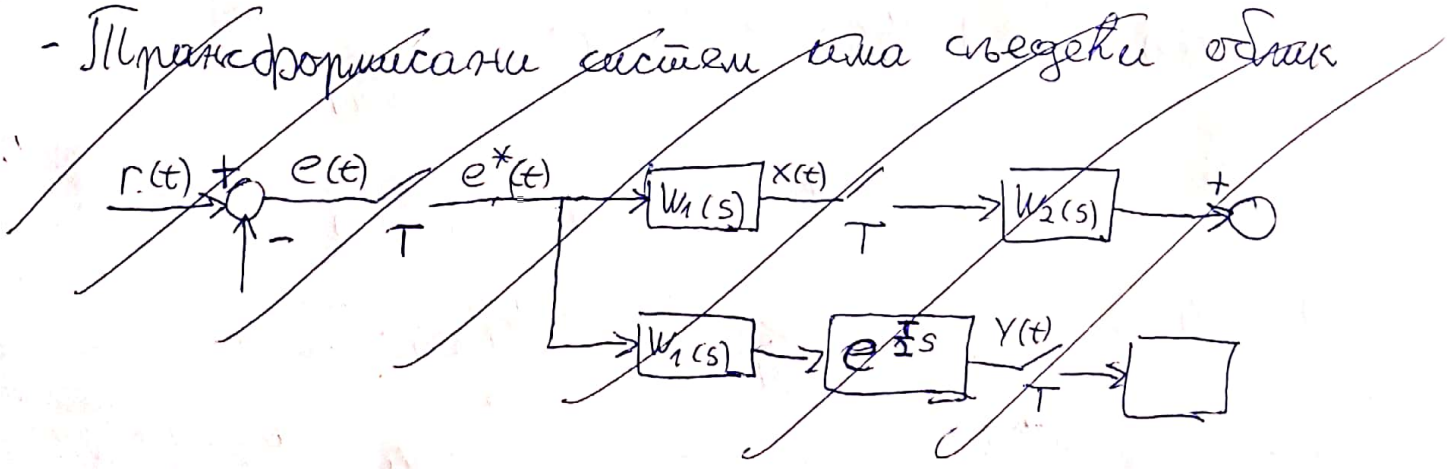
$$W_1(s) = G_{h_0}(s) \cdot G_1(s) = \frac{1 - e^{-sT}}{s} \cdot \frac{1}{s}$$

$$W_2(s) = G_{h_0}(s) \cdot G_2(s) = \frac{1 - e^{-sT}}{s} \cdot \frac{1}{s+1}$$

$$\left.\begin{array}{l} T_1 = 2\,[s] \\ T_2 = 1\,[s] \end{array}\right\} \Rightarrow T_1 = 2T_2$$

$$f^*(t) = f_1^*(t) + f_2^*\left(t + \frac{T}{2}\right)$$

- Трансформисани систем има сљедећи облик



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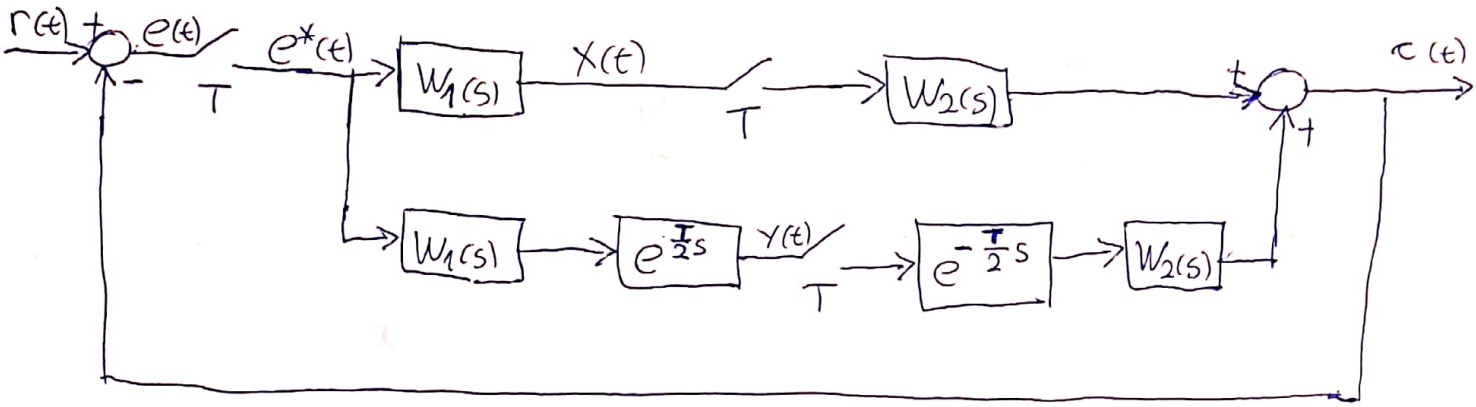
- Трансформисани систем има следећи облик



- У општем случају важи:

$$Z\left\{W_i(s)\cdot e^{-\frac{kT}{n}s}\right\} = Z_m\left\{W_i(s)\right\}_{m=1-\frac{k}{n}} = W_i(z,m)\ ;\ k=0,1,\dots,n-1$$

$$Z\left\{W_i(s)\cdot e^{\frac{kT}{n}s}\right\} = z\cdot Z\left\{W_i(s)\cdot e^{-\left(1-\frac{k}{n}\right)Ts}\right\} = z\cdot Z_m\left\{W_i(s)\right\}_{m=\frac{k}{n}} =$$

$$= z\cdot W_i(z,m)\ ;\ k=0,1,\dots,n-1$$

- Ф-ја спрегнутог преноса у општем случају је

$$G(z) = \frac{C(z)}{R(z)} = \frac{z\cdot\sum_{k=0}^{n-1} W_1\left(z,\frac{k}{n}\right)\cdot W_2\left(z,1-\frac{k}{n}\right)}{1+z\cdot\sum_{k=0}^{n-1} W_1\left(z,\frac{k}{n}\right)\cdot W_2\left(z,1-\frac{k}{n}\right)}$$

- За наш пример имамо

$$C(s) = W_2(s)\cdot X^*(s) + W_2(s)\cdot e^{-\frac{T}{2}s}\cdot Y^*(s) \xRightarrow[\text{ТАФДП}]{}$$

$$C^*(s) = W_2^*(s)\cdot X^*(s) + \left[W_2(s)\,e^{-\frac{T}{2}s}\right]^*\cdot Y^*(s)$$

$$X(s) = W_1(s)\cdot E^*(s) \xRightarrow[\text{ТАФДП}]{} X^*(s) = W_1^*(s)\cdot E^*(s)$$

$$Y(s) = W_1(s)\cdot e^{\frac{T}{2}s}\cdot E^*(s) \xRightarrow[\text{ТАФДП}]{} Y^*(s) = \left[W_1(s)\,e^{\frac{T}{2}s}\right]^*\cdot E^*(s)$$

$$E(s) = R(s) - C(s) \xRightarrow[\text{ТАФДП}]{} E^*(s) = R^*(s) - C^*(s)$$

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$$C^*(s) = W_1^*(s)\cdot W_2^*(s)\cdot E^*(s) + \left[W_2(s)\, e^{-\frac{T}{2}s}\right]^* \cdot \left[W_1(s)\, e^{\frac{T}{2}s}\right]^* \cdot E^*(s)$$

$$C^*(s) = \left[W_1^*(s)\cdot W_2^*(s) + \left[W_1(s)\, e^{\frac{T}{2}s}\right]^* \cdot \left[W_2(s)\, e^{-\frac{T}{2}s}\right]^*\right]\cdot\left[R^*(s) - C^*(s)\right] \Big/ z$$

$$C(z) = \left[W_1(z)\cdot W_2(z) + Z\left\{W_1(s)\, e^{\frac{T}{2}s}\right\}\cdot Z\left\{W_2(s)\, e^{-\frac{T}{2}s}\right\}\right]\left[R(z) - C(z)\right]$$

$$Z\left\{W_1(s)\, e^{\frac{T}{2}s}\right\} = z\cdot W_1\left(z, \tfrac{1}{2}\right)$$

$$Z\left\{W_1(s)\, e^{-\frac{T}{2}s}\right\} = W_2\left(z, \tfrac{1}{2}\right)$$

$$C(z)\left[1 + W_1(z)\cdot W_2(z) + z\, W_1\left(z, \tfrac{1}{2}\right)\cdot W_2\left(z, \tfrac{1}{2}\right)\right] =$$

$$= \left[W_1(z)\cdot W_2(z) + z\, W_1\left(z, \tfrac{1}{2}\right)\cdot W_2\left(z, \tfrac{1}{2}\right)\right] R(z)$$

$$G(z) = \frac{C(z)}{R(z)} = \frac{W_1(z)\cdot W_2(z) + z\, W_1\left(z, \frac{1}{2}\right)\cdot W_2\left(z, \frac{1}{2}\right)}{1 + W_1(z)\cdot W_2(z) + z\, W_1\left(z, \frac{1}{2}\right)\cdot W_2\left(z, \frac{1}{2}\right)}$$

$$W_1(z) = Z\left\{W_1(s)\right\} = Z\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s}\right\} = (1 - z^{-1})\, Z\left\{\frac{1}{s^2}\right\}$$

$$W_1(z) = \frac{z-1}{z}\cdot\frac{Tz}{(z-1)^2} = \frac{z-1}{z}\cdot\frac{2z}{(z-1)^2} \Rightarrow \underline{\underline{W_1(z) = \frac{2}{z-1}}}$$

$T = 2[s]$, бира се сторија периода одабирања,

$$W_2(z) = Z\left\{W_2(s)\right\} = Z\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s+1}\right\} = (1 - z^{-1})\, Z\left\{\frac{1}{s(s+1)}\right\}$$

$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} = \frac{As + A + Bs}{s(s+1)} \Rightarrow \begin{array}{l} A + B = 0 \\ A = 1,\ B = -1 \end{array}$$

$$W_2(z) = \frac{z-1}{z}\, Z\left\{\frac{1}{s} - \frac{1}{s+1}\right\} = \frac{z-1}{z}\left[\frac{z}{z-1} - \frac{z}{z - e^{-T}}\right]$$

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$$W_2(z) = \frac{z-1}{z}\left[\frac{z}{z-1} - \frac{z}{z-e^{-2}}\right] = 1 - \frac{z-1}{z-0.1353}$$

$$W_2(z) = \frac{\cancel{z} - 0.1353 - \cancel{z} + 1}{z-0.1353} \Rightarrow \underline{\underline{W_2(z) = \frac{0.8647}{(z-0.1353)}}}$$

$$W_1(z,m) = Z\left\{W_1(s)\cdot e^{-(1-m)sT}\right\} = Z_m\left\{W_1(s)\right\} =$$

$$W_1(z,m) = Z_m\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s}\right\} = (1-z^{-1})\, Z_m\left\{\frac{1}{s^2}\right\}$$

$$W_1(z,m) = \frac{z-1}{z}\left[\frac{mT}{z-1} + \frac{T}{(z-1)^2}\right] = \frac{mT}{z} + \frac{T}{z(z-1)}$$

$$W_1(z,m) = \frac{mT(z-1)+T}{z(z-1)} \quad ; \quad m=\frac{1}{2},\ T=2\,[s]$$

$$W_1(z,\tfrac{1}{2}) = \frac{\frac{1}{\cancel{2}}\cdot\cancel{2}(z-1)+2}{z(z-1)} \Rightarrow \underline{\underline{W_1(z,\tfrac{1}{2}) = \frac{z+1}{z(z-1)}}}$$

$$W_2(z,m) = Z\left\{W_1(s)\cdot e^{-(1-m)sT}\right\} = Z_m\left\{W_1(s)\right\}$$

$$W_2(z,m) = Z_m\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s+1}\right\} = (1-z^{-1})\, Z_m\left\{\frac{1}{s(s+1)}\right\}$$

$$W_2(z,m) = (1-z^{-1})\, Z_m\left\{\frac{1}{s} - \frac{1}{s+1}\right\} = \frac{z-1}{z}\left[\frac{1}{z-1} - \frac{e^{-mT}}{z-e^{-T}}\right]$$

$$T=2\,[s]\ ,\quad m=\frac{1}{2}$$

$$W_2(z,\tfrac{1}{2}) = \frac{z-1}{z}\left[\frac{1}{z-1} - \frac{e^{-\frac{1}{2}\cdot 2}}{z-e^{-2}}\right] = \frac{z-1}{z}\left[\frac{1}{z-1} - \frac{0.3679}{z-0.1353}\right]$$

$$W_2(z,\tfrac{1}{2}) = \frac{1}{z} - \frac{0.3679(z-1)}{z(z-0.1353)} = \frac{z-0.1353-0.3679z+0.3679}{z(z-0.1353)}$$

$$\underline{\underline{W_2(z,\tfrac{1}{2}) = \frac{0.6321z+0.2326}{z(z-0.1353)}}}$$

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$$G(z) = \frac{W_1(z)\cdot W_2(z) + z W_1(z,\tfrac{1}{2})\cdot W_2(z,\tfrac{1}{2})}{1+W_1(z)\cdot W_2(z) + z\, W_1(z,\tfrac{1}{2})\cdot W_2(z,\tfrac{1}{2})}$$

$$G(z) = \frac{\frac{2}{(z-1)}\cdot\frac{0.8647}{(z-0.1353)} + \cancel{z}\cdot\frac{z+1}{\cancel{z}(z-1)}\cdot\frac{0.6321z+0.2326}{z(z-0.1353)}}{1+\frac{2}{(z-1)}\cdot\frac{0.8647}{(z-0.1353)} + \cancel{z}\cdot\frac{z+1}{\cancel{z}(z-1)}\cdot\frac{0.6321z+0.2326}{z(z-0.1353)}}$$

$$G(z) = \frac{\frac{2z\cdot 0.8647 + (z+1)(0.6321z+0.2326)}{\cancel{z(z-1)(z-0.1353)}}}{\frac{z(z-1)(z-0.1353) + 2z\cdot 0.8647 + (z+1)(0.6321z+0.2326)}{\cancel{z(z-1)(z-0.1353)}}}$$

$$G(z) = \frac{1.7294z + 0.6321z^2 + 0.2326z + 0.6321z + 0.2326}{z^3 - 1.1353z^2 + 0.1353z + 1.7294z + 0.6321z^2 + 0.2326z + 0.6321z + 0.2326}$$

$$G(z) = \frac{0.6321z^2 + 2.5941z + 0.2326}{z^3 - 0.5032z^2 + 2.7294z + 0.2326}$$

$C(z) = G(z)\cdot R(z)$, $R(z) = \frac{z}{z-1}$ – улаз је јединични одскочни сигнал

$$C(z) = \frac{0.6321z^2 + 2.5941z + 0.2326}{z^3 - 0.5032z^2 + 2.7294z + 0.2326}\cdot\frac{z}{z-1}$$

$$C(z) = \frac{0.6321z^3 + 2.5941z^2 + 0.2326z}{z^4 - z^3 - 0.5032z^3 + 0.5032z^2 + 2.7294z^2 - 2.7294z + 0.2326z - 0.2326}$$

$$\boxed{C(z) = \frac{0.6321z^3 + 2.5941z^2 + 0.2326z}{z^4 - 1.5032z^3 + 3.2326z^2 - 2.4968z - 0.2326}}$$

Scanned with Car